# Technical Project Document — Outdoor Conical Matrix Christmas Tree

*Full engineering specification for the 5–6 m conical LED tree: 12 vertical matrix strips, 4,320 total pixels, local-AP control.*

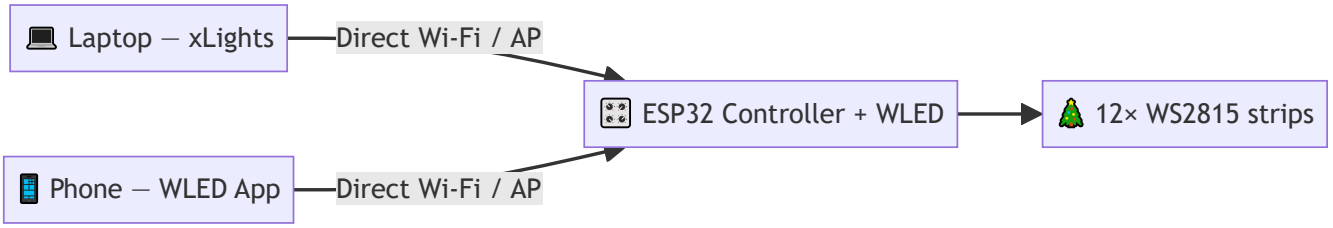
**See also:** README · Single Page Explanation · Software Controller Solution · Setup

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## 1. Executive summary

This document outlines the technical solution for a 5–6 metre outdoor, cone-shaped Christmas tree display. The system treats **12 vertical LED runs as a 3D matrix**, capable of displaying complex, custom-programmed visual effects.

The system features a high-level fail-safe architecture to withstand freezing winter conditions, operates via a direct wireless connection **without relying on home internet infrastructure**, and supports dual control from both a laptop (for advanced sequence design) and a mobile phone (for daily operation).




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## 2. System architecture & software stack

### 2.1 3D pixel mapping

Because a cone narrows sharply at the top, a traditional flat 2D coordinate system will warp or squash spatial effects as they travel up the tree. The solution relies on **3D coordinate mapping**: every individual LED is mapped via explicit X, Y and Z spatial coordinates within the design software, allowing complex effects (rotating spheres, spirals, multi-directional waves) to flow seamlessly across the physical volume of the tree without geometric distortion.

### 2.2 Software stack layout

| Layer | Tool | Runs on | Role |
|---|---|---|---|
| **Design & sequencing** | xLights (open-source) | Laptop | Maps the 12 strips into a 3D conical matrix model; authors timeline effects; audio sync (optional); playlists; effect timing, speeds, loops. |
| **Controller firmware** | WLED (open-source) | ESP32 | Hosts the standalone Wi-Fi AP; decodes live incoming pixel streams from xLights; stores localized presets; exposes a network control API. |
| **Daily operation** | WLED Mobile App / Web UI | Phone | On/off, global brightness, effect speed, select saved sequences or static palettes — without booting the laptop. |

> Detailed software behaviour, sequence delivery, on/off control and one-time commissioning are covered in the Software Controller Solution.

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## 3. Hardware specification

### 3.1 Microcontroller & signal control

- **Brain:** ESP32 development board with an **external SMA antenna port** (rather than an onboard PCB trace antenna), so a high-gain antenna can extend outside the weatherproof enclosure.
- **Interface integration: QuinLED-Dig-Quad** (pre-assembled board v3 or v4), integrating critical protective infrastructure:
  - Onboard **74AHCT125 level shifters** cleanly boost the ESP32's native 3.3 V data signal to the 5 V logic required by the LED strips over long distances.
  - Integrated **safety fuses** for multiple independent power-injection outputs.
- **Performance:** With 4,320 total pixels split across **4 dedicated output channels** (3 strips / 1,080 pixels per channel), the ESP32 achieves a fluid, lag-free **40+ FPS** over a direct Wi-Fi AP link.

### 3.2 High-availability pixels

| Spec | Value |
|---|---|
| Light source | WS2815 12 V DC digital addressable LED strips |
| Density | 60 LEDs/m → 360 LEDs per 6 m strip → 4,320 pixels across 12 lines |
| Waterproofing | IP65 (clear silicone sleeve casing) |

**Critical winterization requirement:** Standard IP65 sleeves are hollow. You must inject marine-grade silicone sealant into both ends of any cut sleeves and seal terminators tightly with dual-wall, glue-lined heat-shrink tubing to prevent freezing condensation from wicking inside the casing.

**Redundancy logic:** The WS2815 chip uses a **dual-signal line** (Data In and Backup Data In). If an individual LED chip fails or cracks due to thermal shock or winter weather, the backup line automatically bypasses it, allowing all subsequent downstream LEDs in that 6 m run to keep operating.

> ⚠️ At the initial input injection point, the **Backup In (BI) pin must be tied directly to Ground (GND)** to initialize the fail-safe channel. See Addendum C.

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## 4. Power infrastructure

WS2815 strips operate on a constant-current architecture, drawing a relatively stable **0.015 A per pixel** across operational states.

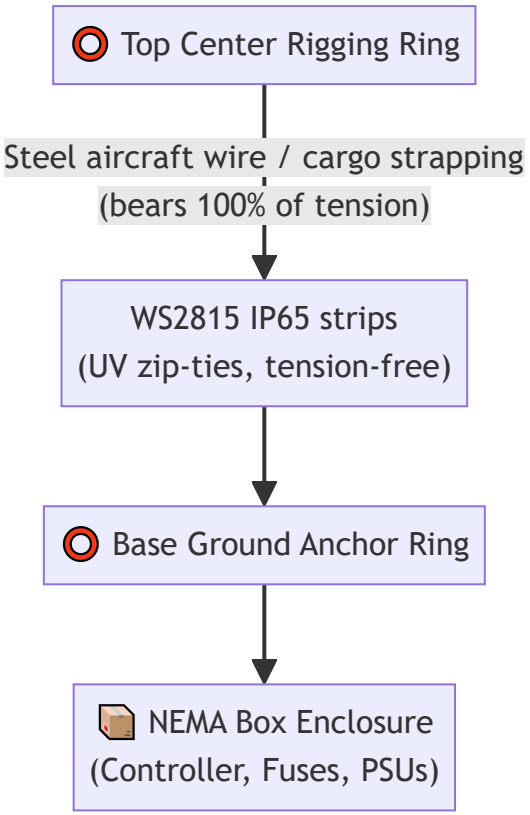
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Calculation:  4,320 pixels × 0.015 A = 64.8 A at 12 V DC
Target continuous wattage:  ≈ 777 W
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| Item | Spec |
|---|---|
| Peak current | 64.8 A @ 12 V |
| Continuous target | ~777 W |
| PSU option A | 2× Mean Well **HLG-600H-12A** (600 W, IP67 outdoor, parallel) |
| PSU option B | 2× Mean Well **LRS-350-12** (350 W indoor, inside a larger ventilated weatherproof box) |
| Recommended margin | ~20% safety overhead |

> The recommended build (see Addendum E) uses 2× Mean Well **HLG-480H-12A** in a split-load configuration.

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## 5. Physical rigging & winterization




### 5.1 Mechanical strain relief

Flexible LED strips have thin copper PCBs that cannot hang freely under their own weight at 6 m; wind loads and freezing ice weight will snap the internal traces. Each of the 12 strips must be securely affixed **every 15–20 cm** to a structural backbone (heavy-duty plastic wire mesh, 2 mm steel aircraft guide wires, or outdoor-rated polyester cargo webbing). **The structural backbone must bear 100% of the physical tension and wind load.**

### 5.2 Data signal integrity

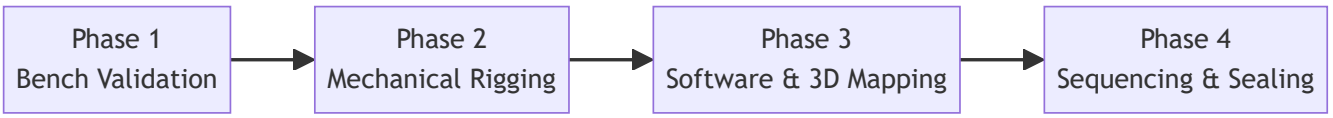
The distance from the ground enclosure to the top center rigging ring may span several metres. Because a 5 V data signal can degrade over long copper extensions, a data booster/amplifier or a **sacrificial single LED pixel** must be placed inline if the distance between the QuinLED board and the first true pixel **exceeds 3 metres**.

### 5.3 Power injection topology

To eliminate voltage drop and the resulting pinkish colour-shift across the 6 m run, 12 V DC power must be fed **simultaneously from both ends** of the loops. Run heavy-gauge (14 AWG) weather-insulated power buses up the center trunk to feed the top-ring injection points, and split parallel lines across the bottom-ring anchors.

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## 6. Project implementation phasing




**Phase 1 — Bench validation (desk scale)**

- Flash WLED firmware onto the ESP32 board.
- Configure standalone Wi-Fi AP credentials and test external antenna signal range.
- Wire a single 6 m test strip to the QuinLED output to verify backup data routing and monitor continuous heat dissipation.

**Phase 2 — Mechanical rigging & assembly**

- Erect the main center mast; anchor the top center ring and wide base ground ring.
- Run the 12 structural backing lines tautly between the rings.
- Secure the 12 WS2815 strips to the lines with heavy-duty, UV-stabilized zip-ties.
- Route the top and bottom 14 AWG power-injection buses; seal all outdoor joints.

**Phase 3 — Software & 3D pixel mapping**

- Boot the controller into AP mode; connect the design laptop to the local network.
- In xLights create a Custom Model / Cone layout element.
- Input the exact physical pixel routing order (e.g. String 1: top-to-bottom, String 2: bottom-to-top) into the xLights spatial mapper to build an accurate 3D preview matrix.

**Phase 4 — Sequence design & weatherproof enclosure seals**

- Build custom effects, sequences and timing loops on the xLights timeline.
- For standalone seasonal playback, export the compiled effects and save them as presets on the controller's onboard storage.
- For live real-time control/editing, stream data from the laptop using E1.31 or DDP over the local Wi-Fi link.
- Encase all base electronics in the NEMA box, secure liquid-tight cable glands, and position the external high-gain antenna away from metal framing.

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# Addenda


- Addendum A — IP Rating Comparison
- Addendum B — WS2811 vs WS2815
- Addendum C — WS2815 ↔ QuinLED-Dig-Quad Wiring
- Addendum D — 4-Channel Strip Layout & WLED Config
- Addendum E — Power Architecture & Enclosure Mounting

- Addendum F — Bill of Materials

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## Addendum A — IP Rating Comparison

| Rating | Solids (1st digit) | Liquids (2nd digit) | Submersion capable? | Best used for |
|---|---|---|---|---|
| **IP65** | 6 — Dust-tight | 5 — Water jets | No | Outdoor fixtures, washdown environments |
| **IP67** | 6 — Dust-tight | 7 — Immersion | Yes (up to 30 min) | Rain exposure, accidental drops, temporary puddles |
| **IP68** | 6 — Dust-tight | 8 — Continuous | Yes (continuous) | Pool lights, dive gear, premium smartphones |

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## Addendum B — WS2811 vs WS2815 Technical Comparison

### Summary table

| Attribute | WS2811 (5 V) | WS2815 (12 V) |
|---|---|---|
| Voltage | 5 V | 12 V |
| Current per pixel | up to 60 mA at full white | ~60 mA internally, far lower current **per metre** (LEDs powered in groups) |
| Voltage drop | Significant after 50–100 px | Far less severe |
| Power-injection spacing | Every 50–100 px | Every 150–300 px (density dependent) |
| Failure mode | One dead pixel → **whole downstream chain goes dark** | Dual-signal bypass → **only the failed pixel goes dark** |
| Signal protocol | 800 kHz NRZ | 800 kHz NRZ (**identical**) |
| Wiring | Heavy copper (high current) | Thinner wire OK (lower current) |
| Cost & availability | Cheaper, very common | More expensive, increasingly the professional default |

### 1. Electrical architecture

- **WS2811 (5 V):** each pixel draws up to 60 mA at full white; voltage drop becomes significant after 50–100 pixels; requires frequent power injection; wiring must handle high current.

- **WS2815 (12 V):** each pixel still draws ~60 mA internally, but the strip draws much less current per metre because LEDs are powered in groups; voltage drop is far less severe; injection spacing is significantly longer.
- **Practical effect:** WS2815 is far easier to power cleanly in large installations.

## 2. Failure modes

- **WS2811:** if one pixel dies, the entire chain downstream goes dark — the single biggest operational weakness.
- **WS2815:** dual-signal / bypass design — if a pixel dies, the data line continues to the next pixel; only the failed pixel goes dark.
- **Practical effect:** WS2815 is dramatically more reliable for permanent or semi-permanent installations.

## 3. Signal & timing

Both use the same **800 kHz NRZ** protocol, meaning:

- The ESP32 RMT driver works identically.
- Frame-slicing / compression pipeline is unchanged.
- The coordinate mapper and global indexer don't care which chip is used.
- **No firmware changes required** to switch chips.

## 4. Power budgeting

- **WS2811:** 5 V × high current = heavy copper; injection every 50–100 px; sensitive to voltage sag → colour shifts.
- **WS2815:** 12 V distribution → lower current → thinner wires OK; injection every 150–300 px; much more tolerant of long runs; reduces brownout risk and simplifies power-budget enforcement.

## 5. Cost & availability

WS2811 is cheaper and extremely common. WS2815 is more expensive but increasingly the professional default. Given this project's scale and reliability goals, **WS2815 is the better long-term choice.**

## 6. Verdict

WS2811 is fine for prototyping and small matrices. **WS2815 becomes compelling — and for a large installation is the clear winner — when you scale beyond short runs, want fewer injection points, need resilience against pixel failure, and want cleaner wiring with lower current.**

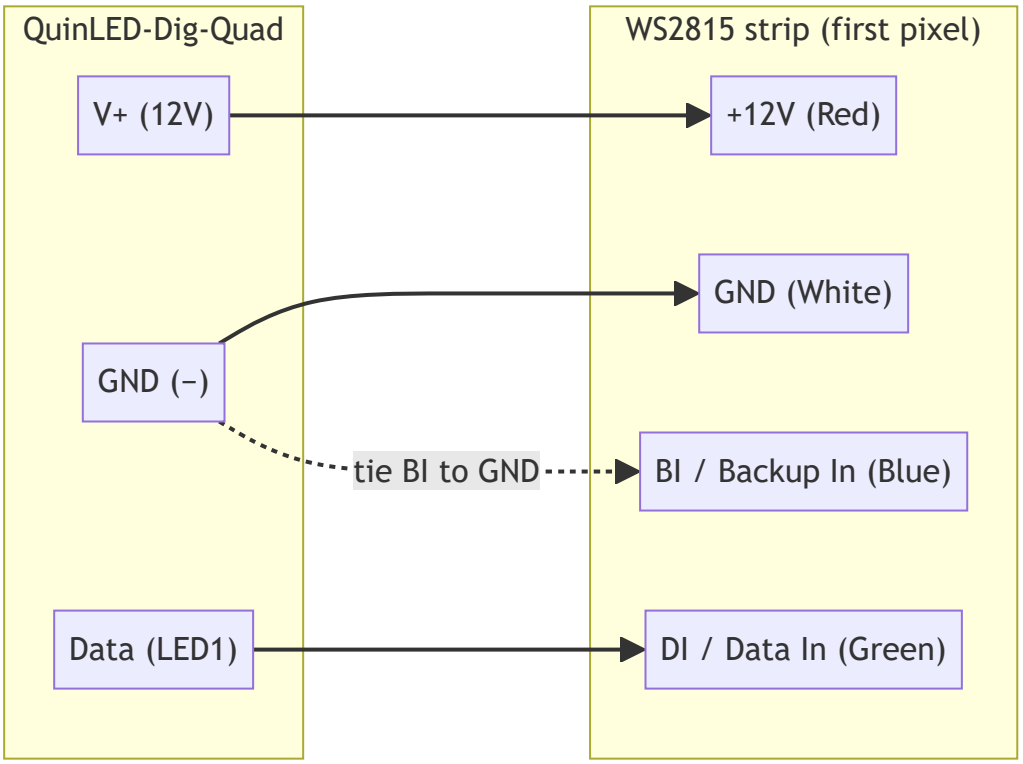
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# Addendum C — WS2815 ↔ QuinLED-Dig-Quad Wiring & Integration

The Dig-Quad provides dedicated terminals for Positive (+), Negative (−) and Data (D). A weatherproof WS2815 strip arrives with 4 incoming wires (typically Red, White, Green, Blue).

## Terminal-to-wire map

| QuinLED-Dig-Quad terminal | WS2815 wire | Notes |
|---|---|---|
| **V+ (12 V)** | Red (+12 V) | Power |
| **GND (−)** | White (GND) | Ground |
| **Data (e.g. LED1)** | Green (DI / Data In) | Onboard 74AHCT125 level shifter drives this to 5 V |
| **GND (shared)** | Blue (BI / Backup In) | **Tie Backup In to Ground at the first injection point** |




## Three critical rules

1. **At the start of each run** — tie **BI** → **GND** (e.g. join White + Blue into the same WAGO before the controller). The chip manufacturer specifies the first BI pin must read a "0" so the first pixel initiates backup logic. **Do not** tie BI to the primary Data line (causes flickering / signal noise).
2. **At intermediate extension joints** — solder all 4 pads sequentially (+12 V→+12 V, GND→GND, DO→DI, **BO→BI**). Do **not** ground the backup line mid-run: the Backup Output (BO) of pixel #300 must connect straight to the Backup Input (BI) of pixel #301.

3. **At cut silicone sleeves** — fill cut ends with **marine-grade clear electronics silicone** (non-corrosive / alkoxy / neutral cure) and shrink **dual-wall adhesive-lined heat-shrink** over the entire seam to keep out condensation through temperature cycling.

### QuinLED-Dig-Quad advantages for this setup

- **Auto-voltage switching** — no manual jumpers; connect a 12 V supply to the main input and the board accepts 12 V, powers the onboard ESP32 safely, and distributes 12 V to the fuse channels.
- **Onboard fuses** — each output channel is protected by a swappable 5 A fuse for short-circuit protection in wet winter weather.
- **Clean 5 V data signals** — native 3.3 V signals often drop out after 2–3 m; the 74AHCT125 level shifters boost data to 5 V so wires reach the top of the 6 m tree without external boosters.

> **Length note:** the design uses 6 m vertical runs but strips are sold in 5 m rolls, so you will cut and solder a 1 m extension onto each run — apply rules 2 and 3 at every joint.

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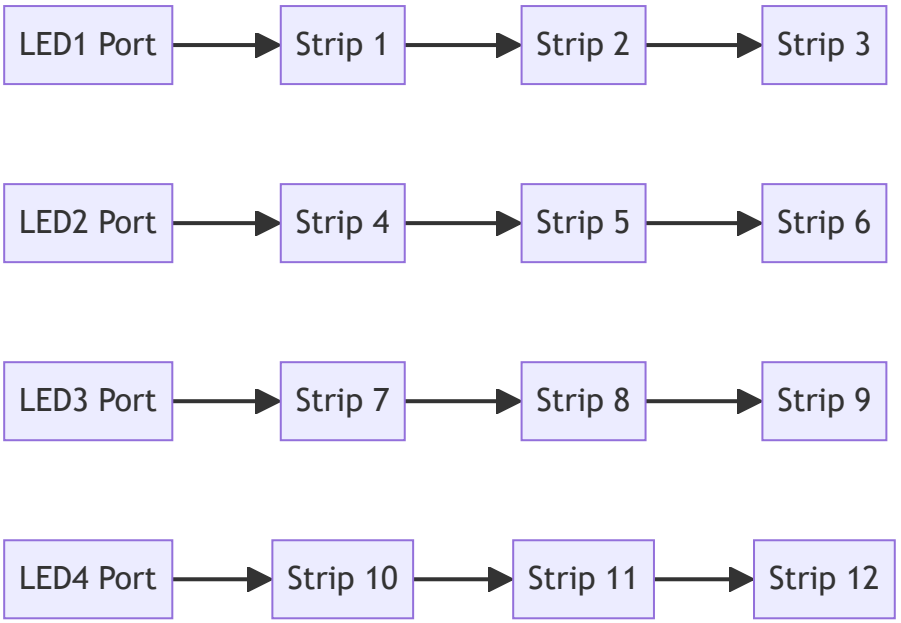
## Addendum D — 4-Channel Strip Layout & WLED Configuration

A single QuinLED-Dig-Quad running WLED controls all 12 strips. The board has **4 dedicated data outputs** (LED1–LED4). For best performance, split the 12 strips evenly — **3 strips per output**.

### Optimal 4-channel layout

| Port | Quadrant | Strips | Pixels |
|---|---|---|---|
| LED1 | North | 1–3 | 1,080 |
| LED2 | East | 4–6 | 1,080 |
| LED3 | South | 7–9 | 1,080 |
| LED4 | West | 10–12 | 1,080 |
| **Total** | — | **12** | **4,320** |

*360 pixels per strip (60 LEDs/m × 6 m); 3 strips end-to-end per port = 1,080 px/port.*




### Why this configuration

1. **Maximises frame rate.** Microcontrollers send pixel data sequentially. All 4,320 px on one pin → ~15–20 FPS (choppy). Splitting across 4 ports lets the ESP32 transmit down all 4 paths simultaneously → **42+ FPS** (smooth).
2. **Simplifies routing.** Group outputs by quadrant (North/East/South/West) so data wires stay cleanly separated around the base instead of one massive chain.
3. **Eases power injection.** The Dig-Quad has 7 positive and 7 negative fused power terminals. Use 4 to feed the "Data In" sides at the bottom of each quadrant; use the remaining 3 to run heavy-gauge lines up the trunk to inject power at the top rings.

### WLED output configuration

| Output pin | LED type | Start pixel | Count |
|---|---|---|---|
| GPIO 16 (LED1) | WS281x / WS2815 | 0 | 1,080 |
| GPIO 3 (LED2) | WS281x / WS2815 | 1,080 | 1,080 |
| GPIO 1 (LED3) | WS281x / WS2815 | 2,160 | 1,080 |
| GPIO 4 (LED4) | WS281x / WS2815 | 3,240 | 1,080 |

> WLED reads the WS2815 protocol natively using the standard **WS281x** setting at 800 kHz.

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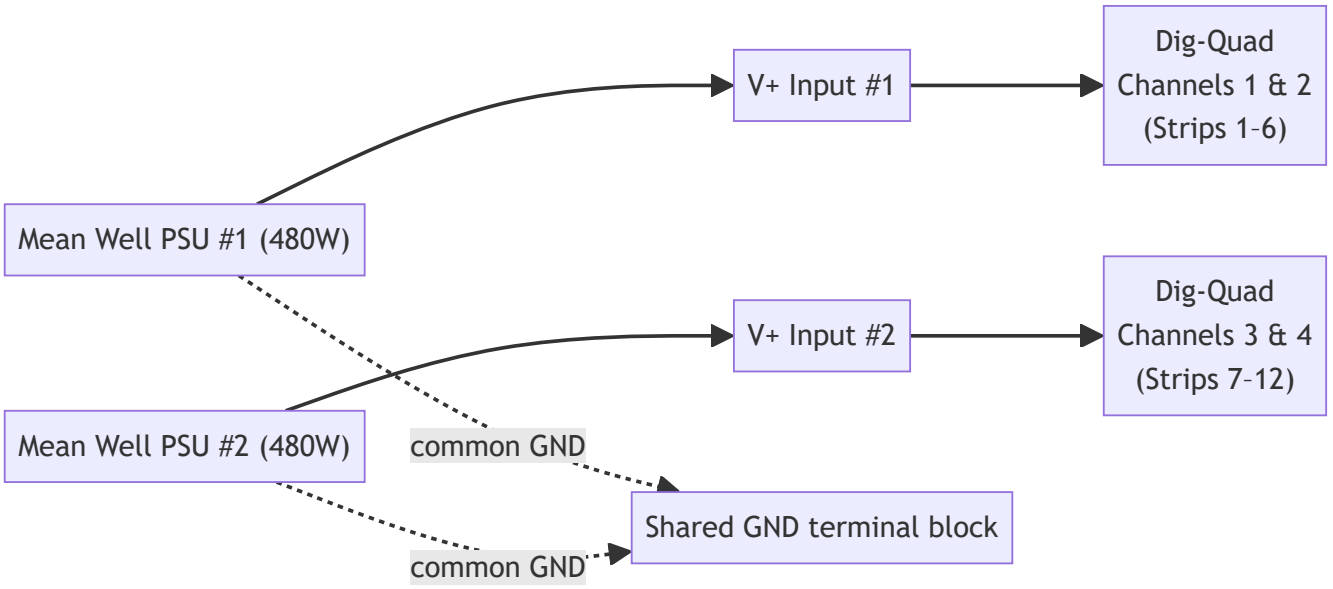
# Addendum E — Power Architecture & Enclosure Mounting

## Part 1 — Powering the system

Because WS2815 strips use a constant-current architecture, they draw a steady load even on dim colours. Managing the ~780 W total safely requires a dual-supply split-load layout.

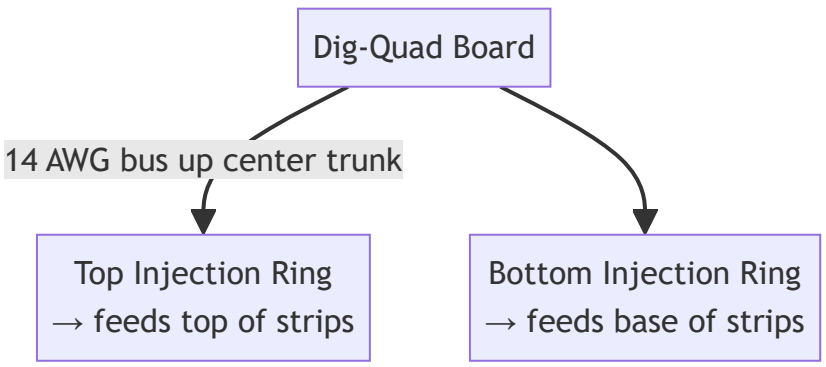
**Dual power supply setup.** Do **not** use a single ~800 W supply (they usually need noisy fans that pull damp winter air into the box). Use two fanless, waterproof supplies in a split-load configuration.

- **Hardware:** 2× **Mean Well HLG-480H-12A** (12 V, 480 W, IP67 waterproof).
- **Split-load rule:** keep the two PSU circuits completely separate. **Never connect the V+ outputs of two different power supplies together.**




- Connect PSU #1 to the first positive input bank → powers Channels 1 & 2.
- Connect PSU #2 to the second positive input bank → powers Channels 3 & 4.
- Tie **all** negative/ground wires from both PSUs together on the common ground terminal block to establish a unified signal reference.

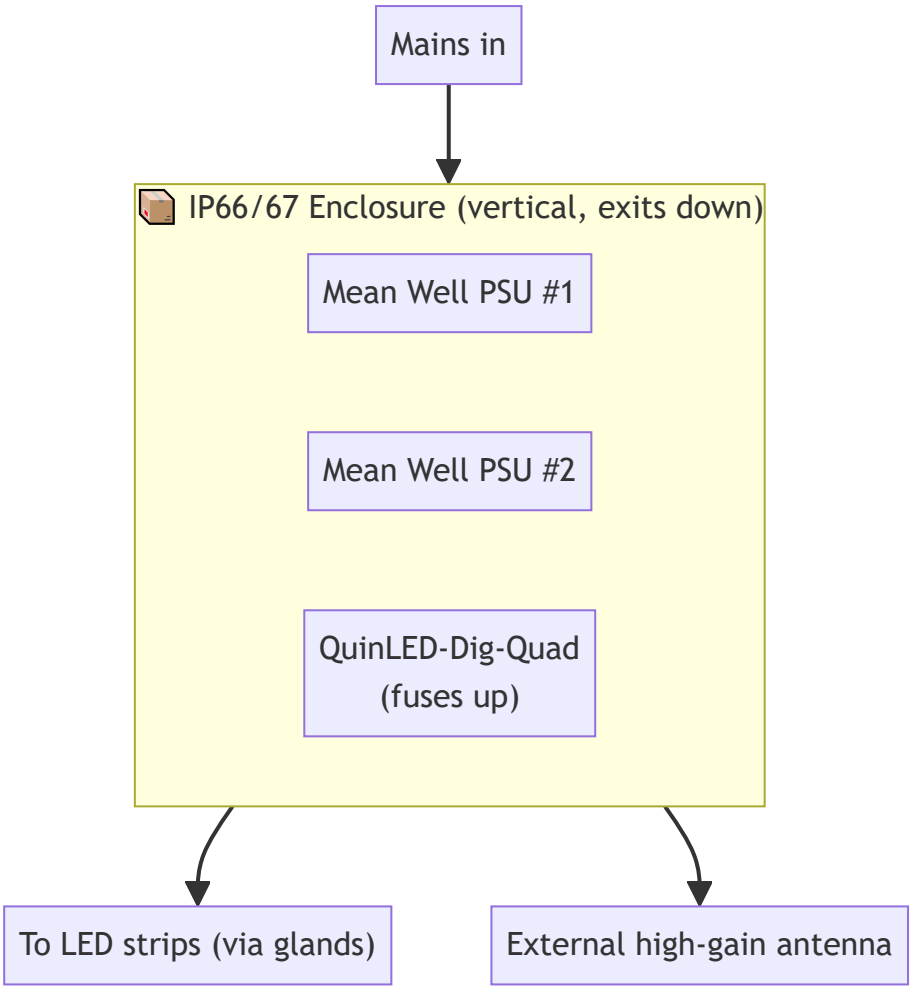
**Power injection plan.** A 12 V signal down a 6 m strip drops in voltage, dimming the far end to a dull pinkish-red. Feed power from **both ends** of each 3-strip chain:



- **At the base:** connect 12 V + GND from the Dig-Quad to the start of the first strip in each quadrant.
- **At the top:** run dedicated heavy-gauge 14 AWG weather-insulated wire up the center trunk to a top-ring distribution node; connect to the end of the third strip in each chain.
- **Fusing:** protect every injection branch with a 5 A or 7.5 A automotive spade fuse on the Dig-Quad.

## Part 2 — Mounting in an IP66/IP67 enclosure

| Requirement | Specification |
|---|---|
| Rating | IP66 or IP67 (dust-tight; powerful water jets / temporary submersion) |
| Material | Polycarbonate or heavy ABS — **never metal** (Faraday cage blocks the Wi-Fi AP) |
| Size | ~300 × 250 × 150 mm, hinged latched lid, integrated rubber gasket |
| Cable exits | **Bottom wall only** — creates a natural drip loop |
| Cable glands | Nylon IP68 compression fittings (PG7/PG9) cranked onto each cable jacket |
| Antenna | SMA-to-IPEX pigtail through the wall → external omnidirectional high-gain antenna pointing up; seal threads with silicone grease |
| Mounting plate | Internal non-conductive boss plate; short self-tappers — never screw into the outer shell |
| Air spacing | 2–3 cm between PSUs and Dig-Quad for passive convection cooling |




- Even though the Mean Well PSUs are IP67 and could sit outside, keeping them inside shields their terminals from ice build-up — but ~800 W of equipment generates heat, hence the air gap and convection spacing.
- **Orientation:** mount the box vertically with wire exits pointing straight down so rainwater runs away from seal points.

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## Addendum F — Bill of Materials

Itemized BOM for the 4,320-LED tree, optimized for electrical safety, winter weatherization and signal integrity.

## 1. Core electronics & control

| Item | Qty | Specification | Purpose |
|---|---|---|---|
| QuinLED-Dig-Quad | 1 | Pre-assembled v3/v4 board (with ESP32 pocket) | 4 channels, level-shifts data to 5 V, holds fuses |
| ESP32 external-antenna board | 1 | ESP32-WROOM-32E with external SMA connector | Core microcontroller; broadcasts standalone Wi-Fi AP |
| High-gain external Wi-Fi antenna | 1 | 2.4 GHz omnidirectional outdoor antenna + SMA extension | Mounts on box exterior to keep a solid link to laptop/phone |
| Automotive spade fuses (5 A & 7.5 A) | 1 pack | Standard mini automotive spade fuses | Populate the Dig-Quad protection slots |

## 2. Fault-tolerant LED pixels

| Item | Qty | Specification | Purpose |
|---|---|---|---|
| WS2815 12 V LED strips (IP65) | 72 m | 60 LEDs/m, 12 V, dual-data backup (buy 15× 5 m rolls) | 12 vertical 6 m runs (4,320 px, includes spares) |
| Marine-grade clear silicone | 1 tube | Non-corrosive, alkoxy/neutral-cure electronics sealant | Inject into cut IP65 sleeve ends to stop condensation |
| Dual-wall heat-shrink tubing | 1 pack | 4:1 or 3:1, marine-grade with internal hot-melt adhesive | Watertight seal over solder joints & end caps |

## 3. Power infrastructure

| Item | Qty | Specification | Purpose |
|---|---|---|---|
| Mean Well HLG-480H-12A | 2 | 12 V DC, 480 W, IP67, fanless outdoor PSU | Clean continuous power, no internal fans |
| 14 AWG 2-core landscape wire | 50 m+ | UV-resistant, stranded tinned-copper outdoor wire | Main power runs from base enclosure to top ring |
| 3-pin waterproof connectors | 12 pairs | IP68 xConnect / Ray Wu pre-moulded pigtails | Quick waterproof strip connection at the base ring |
| WAGO 221 lever connectors | 1 box | 3-wire & 5-wire splicing connectors (keep inside enclosure) | Safely tie grounds & power buses |

## 4. Outdoor weatherproof enclosure

| Item | Qty | Specification | Purpose |
|---|---|---|---|
| IP66/IP67 NEMA enclosure | 1 | Polycarbonate box (~300×250×150 mm), hinged lid | Houses PSUs, Dig-Quad, internal wiring |
| Nylon cable glands (PG9/PG11) | 1 pack | Liquid-tight strain-relief compression fittings | Seal wire-exit holes in the box bottom |
| Outdoor mains power cord | 1 | Heavy-duty 3-core rubberized mains lead (IP44+ plug) | Feed safe local mains to the PSUs |

## 5. Mechanical frame & support (anti-strain)

| Item | Qty | Specification | Purpose |
|---|---|---|---|
| Heavy-duty webbing / mesh | 12 runs | 6 m UV-stabilized plastic mesh strips | Load-bearing backbone supporting strip weight |
| UV-resistant black zip-ties | 1 pack | Outdoor-rated, high-tensile nylon | Secure IP65 sleeves to mesh every 15 cm |
| Steel top ring & anchors | 1 set | Custom steel hoop for top rigging + heavy ground pegs | Structural framework to mount & tension strips |

## Wire-gauge (AWG) safety guardrails

This project pulls up to ~65 A globally — correct gauges prevent fire hazards and voltage drop.

| Run | Gauge | Why |
|---|---|---|
| PSU → Dig-Quad input | **10 or 12 AWG** (short) | Must handle the full 32.4 A out of each Mean Well PSU |
| Dig-Quad → top injection ring | **14 AWG** | Over a 6 m climb, thinner wire (e.g. 18 AWG) drops too much voltage and runs warm |
| Dig-Quad → base strips | **18 AWG** pigtails | OK provided box-to-first-pixel distance is under 2 m |